Data Analytics

Retail Sales Analysis using SQL

Uncovering actionable business insights through structured data cleaning, trend analysis, and category performance evaluation.

```sql
SELECT category,
  COUNT(order_id) AS total_orders,
  SUM(total_sale) AS revenue
FROM retail_sales
WHERE status = 'Completed'
GROUP BY category
ORDER BY revenue DESC;

-- Analysis complete
```

SQL | Data Analysis | Retail Analytics

HR
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Problem Statement

Retail businesses generate vast amounts of raw sales data but often struggle to translate this information into actionable strategies.

Category Performance

Difficulty in isolating best and worst performing product lines amidst large datasets.

Customer Identification

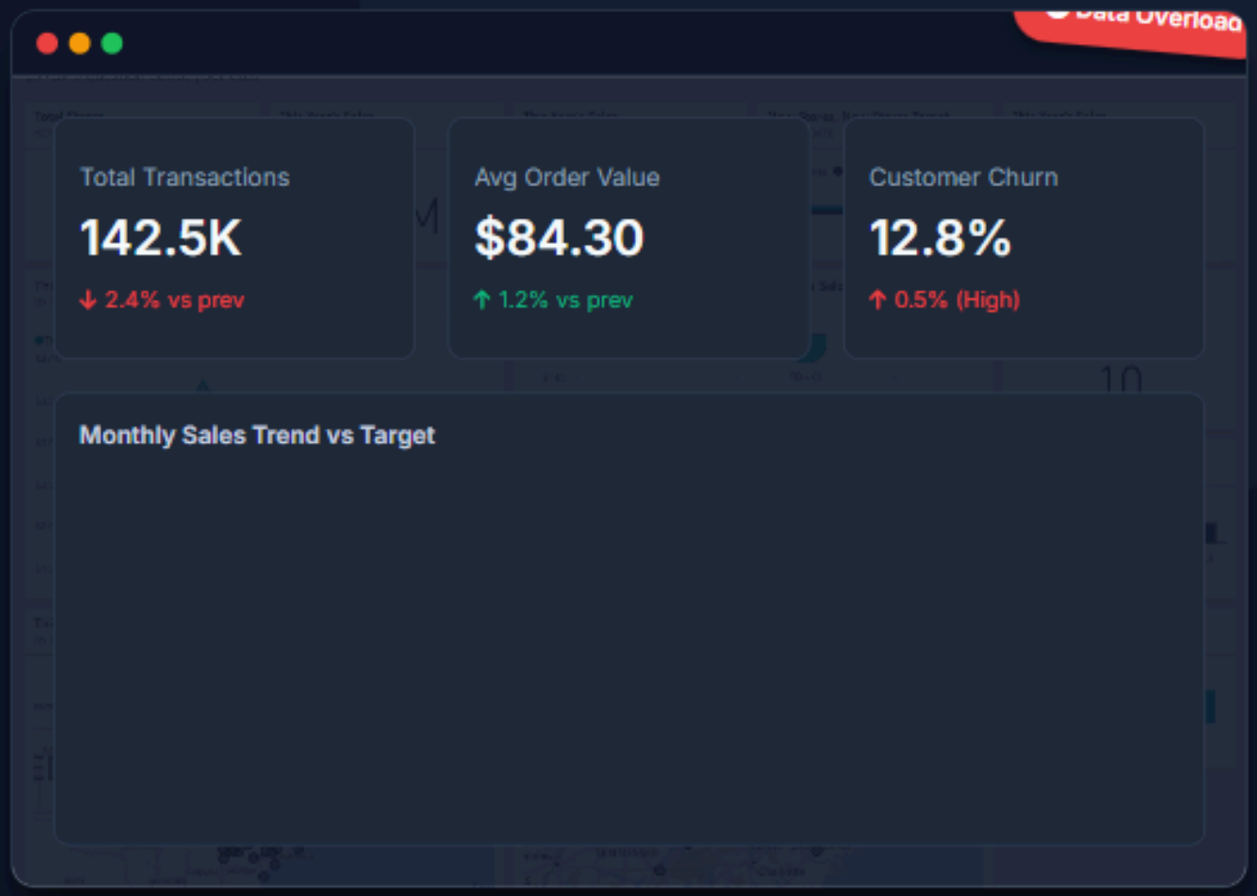
Inability to pinpoint high-value customers (RFM) to target retention efforts effectively.

Sales Trend Tracking

Missed opportunities due to lack of visibility into seasonal trends and peak sales periods.

Buying Behavior Analysis

Understanding time-of-day purchasing patterns (Morning vs. Evening shifts) to optimize staffing.



Solution Approach

End-to-End Structured Data Analysis Pipeline

1 Ingest

Imported raw datasets: Orders, Customers, Products, and Payments tables into SQL environment.

2 Clean

Handled NULL values, standardized date formats, and removed duplicate transaction records.

3 Model

Established relationships (JOINS) and defined key metrics like AOV and Total Revenue.

4 Analyze

Executed queries for Category Performance, Customer Segmentation, and Monthly Trends.

```
-- 4. Trend Analysis Query
SELECT TO_CHAR(sale_date, 'YYYY-MM') as month,
SUM(total_sales) as revenue, COUNT(transaction_id)
as orders
FROM retail_sales
GROUP BY 1
ORDER BY 1 DESC;
```

5 Present

Synthesized findings into actionable business insights regarding sales shifts.

Key Solutions & Insights

Comprehensive Analysis of Retail Data Points

Data Cleaning

Removed null records and standardized data formats to ensure analysis accuracy and reliability.

NULL Removal | De-duplication

Category Analysis

Evaluated total sales volume and order frequency across varying product categories.

Customer Insights

Identified high-value customers and top spenders to drive targeted retention strategies.

Top 100 Identified

Trend Analysis

Analyzed monthly sales trends to identify best-performing periods and seasonality peaks.

Peak Growth

+18.5%

Time-Based Analysis

Studied order patterns by Morning, Afternoon, and Evening shifts to optimize staffing.

AM PM Eve

Skills Demonstrated

Technical Competencies & Analytical Methodologies

```
SELECT skills, proficiency
FROM portfolio_projects
WHERE project_type = 'Retail Analytics'
AND outcome = 'Actionable Insights'
DESC;
```

SQL

Advanced querying using Joins, CTEs, Window Functions, and Aggregations to manipulate large datasets.

PostgreSQL | Complex Queries | Schema Design

Data Cleaning

Handling NULL values, de-duplicating records, and standardizing formats for data integrity.

Validation | Transformation | Quality Check

Exploratory Data Analysis

Uncovering underlying patterns, distributions, and anomalies to guide deeper investigation.

Trend Identification | Descriptive Stats

Retail Analytics

Applying domain knowledge to metrics like AOV, Customer Lifetime Value, and Shift Analysis.

Business Problem Solving

Translating raw data findings into strategic recommendations for inventory and marketing teams.

Thank You!

I hope this presentation provided clear insights into how SQL and data analytics can solve real-world retail challenges.

HR

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